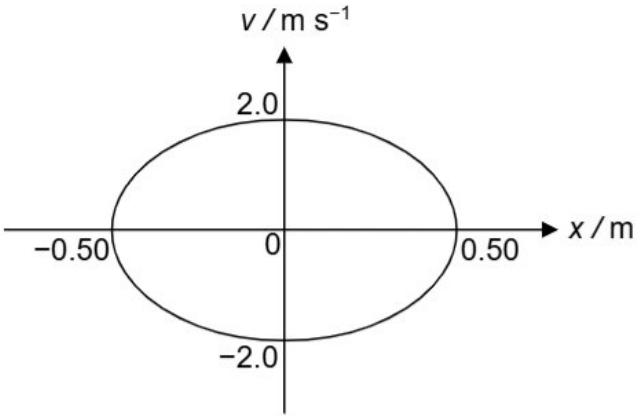
13 A body oscillates in simple harmonic motion. The graph shows the variation with displacement x of the velocity v of the body.



The body is now made to oscillate such that its amplitude is reduced to 0.25 m, while maintaining the same total mechanical energy.

What is the new maximum acceleration of the body?

A 4.0 m s^{-2} **B** 8.0 m s^{-2} **C** 16 m s^{-2} **D** 32 m s^{-2}